Binary Trees

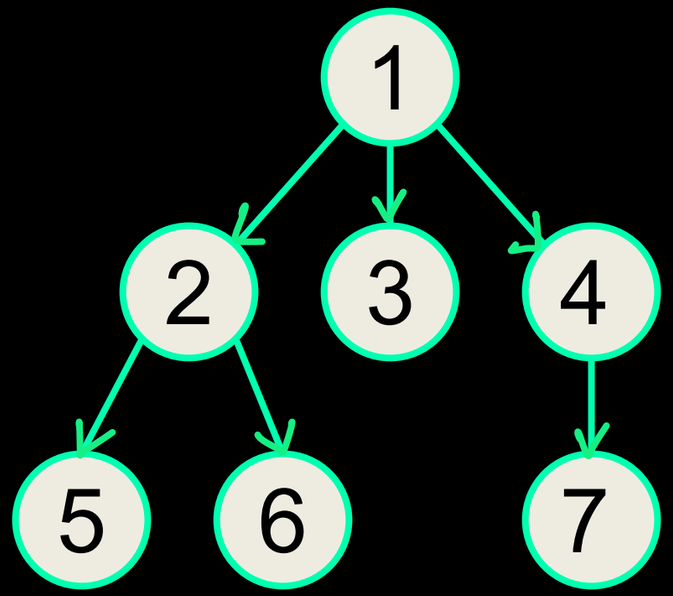
Prerequisites for Trees

1) Recursion – EK method khudko 2 baar agar call kare
preinpost

```
fun() {
    base case
    Kaam
    call
    Kaam
    call
    Kaam
}
```

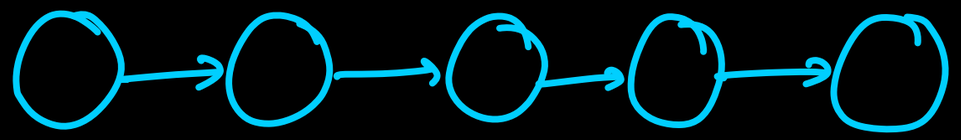
2) Linked List – Node class, null

What is a Tree? 'Heirarchical D.S.' (Generic Tree)

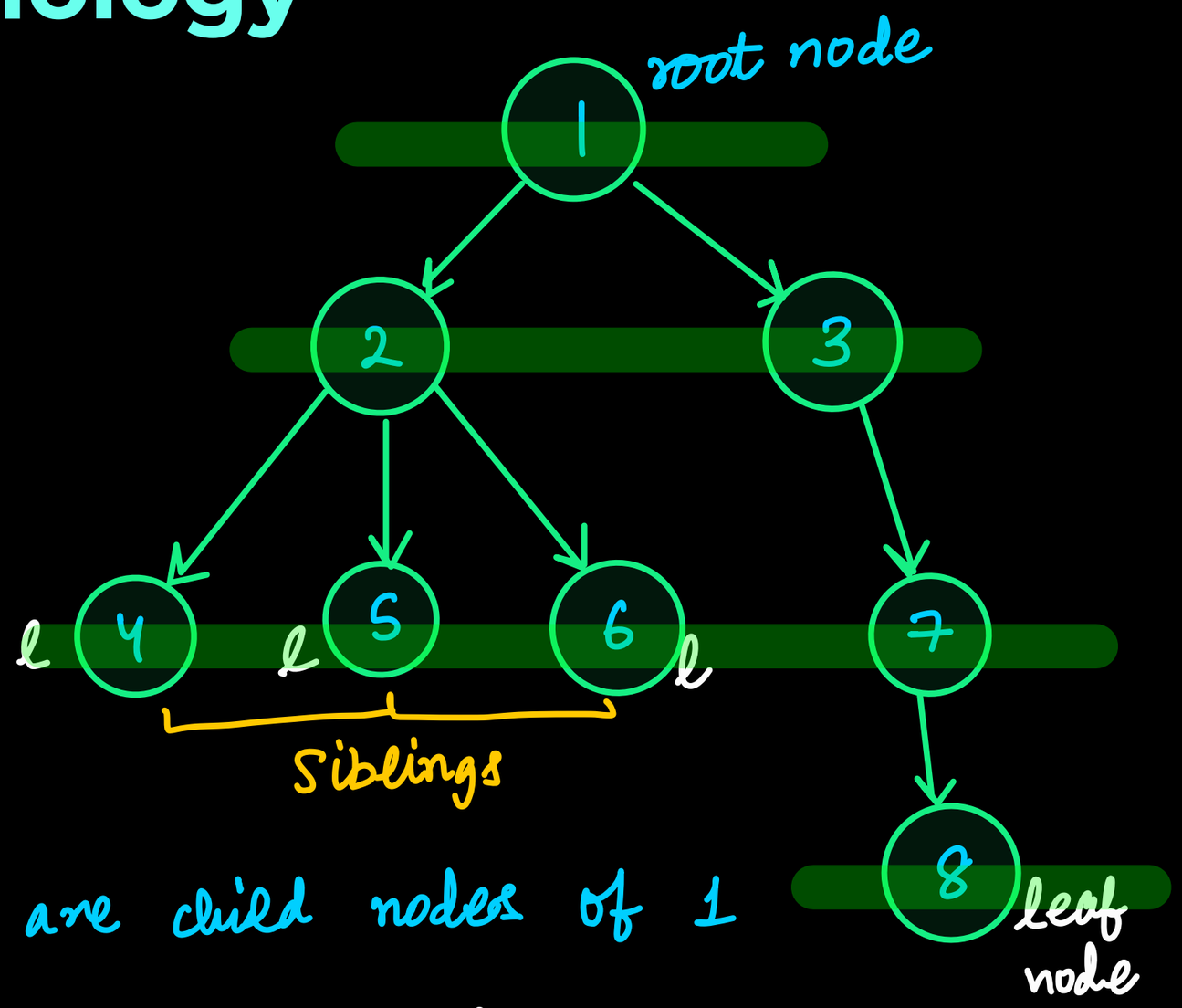


multiple nexts

Linked list :



Terminology



Size = no. of nodes

levels = 4

height = levels - 1

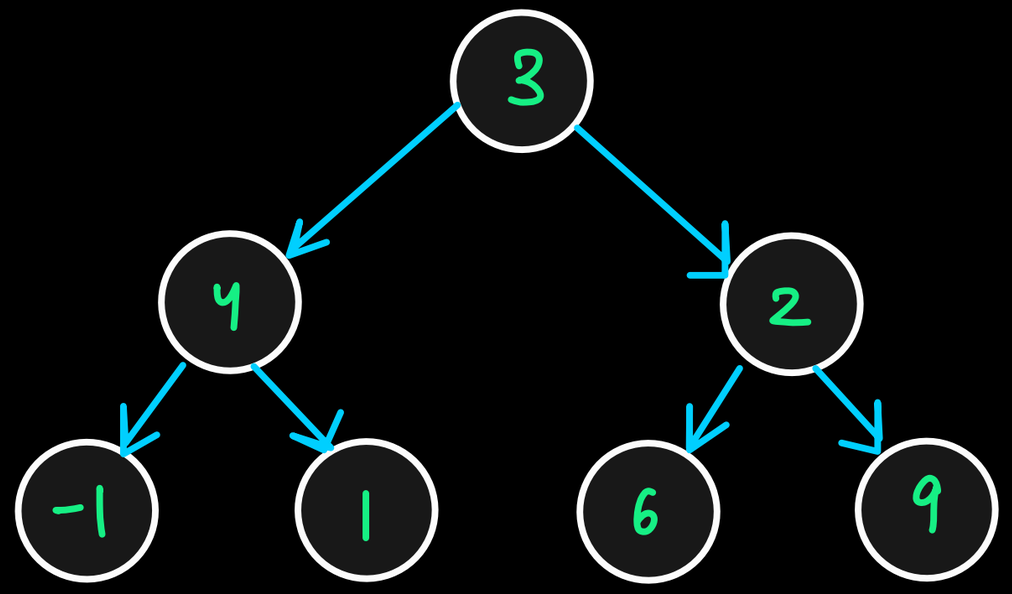
edges = size - 1

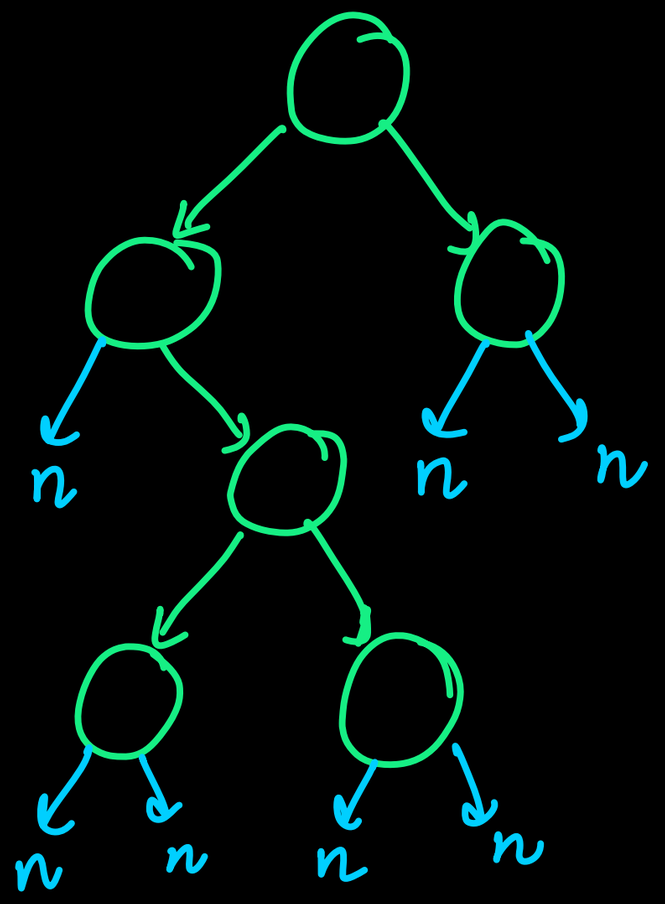
2 & 3 are child nodes of 1

2 is ancestor of 4, 5, 6

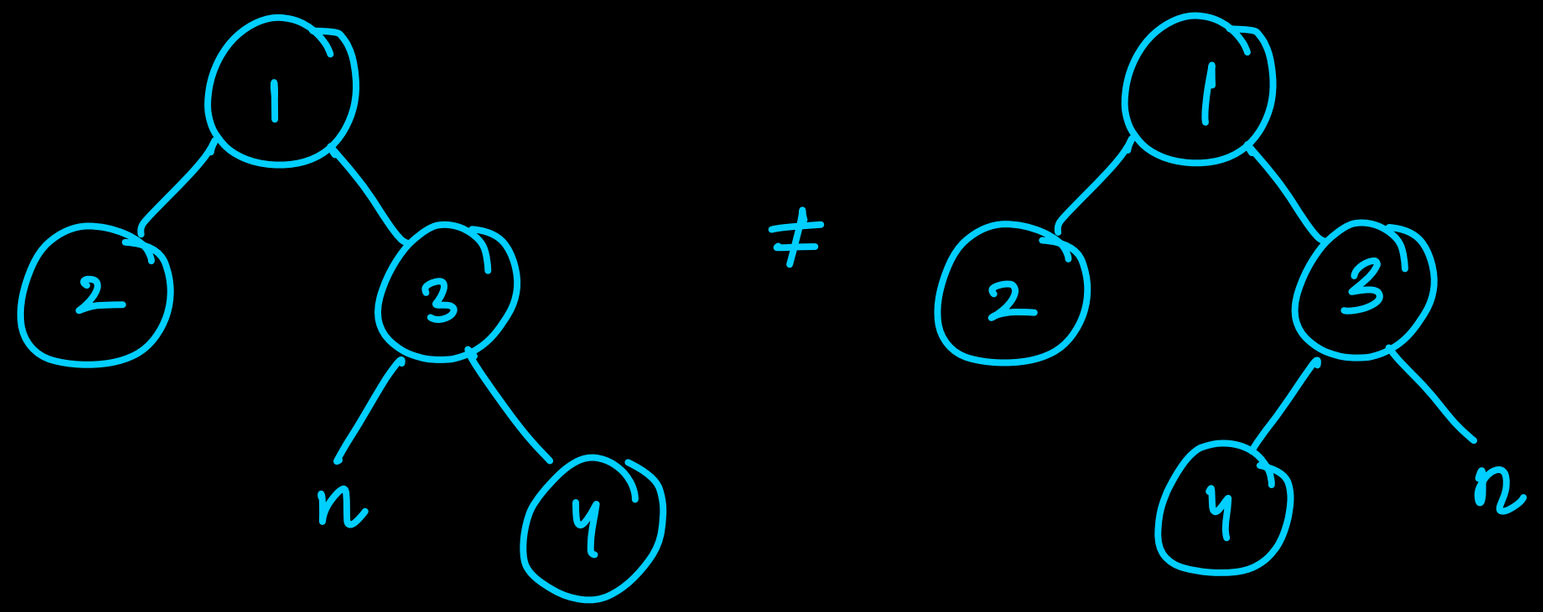
What is a Binary Tree?

Any node can have 0, 1, 2 child nodes





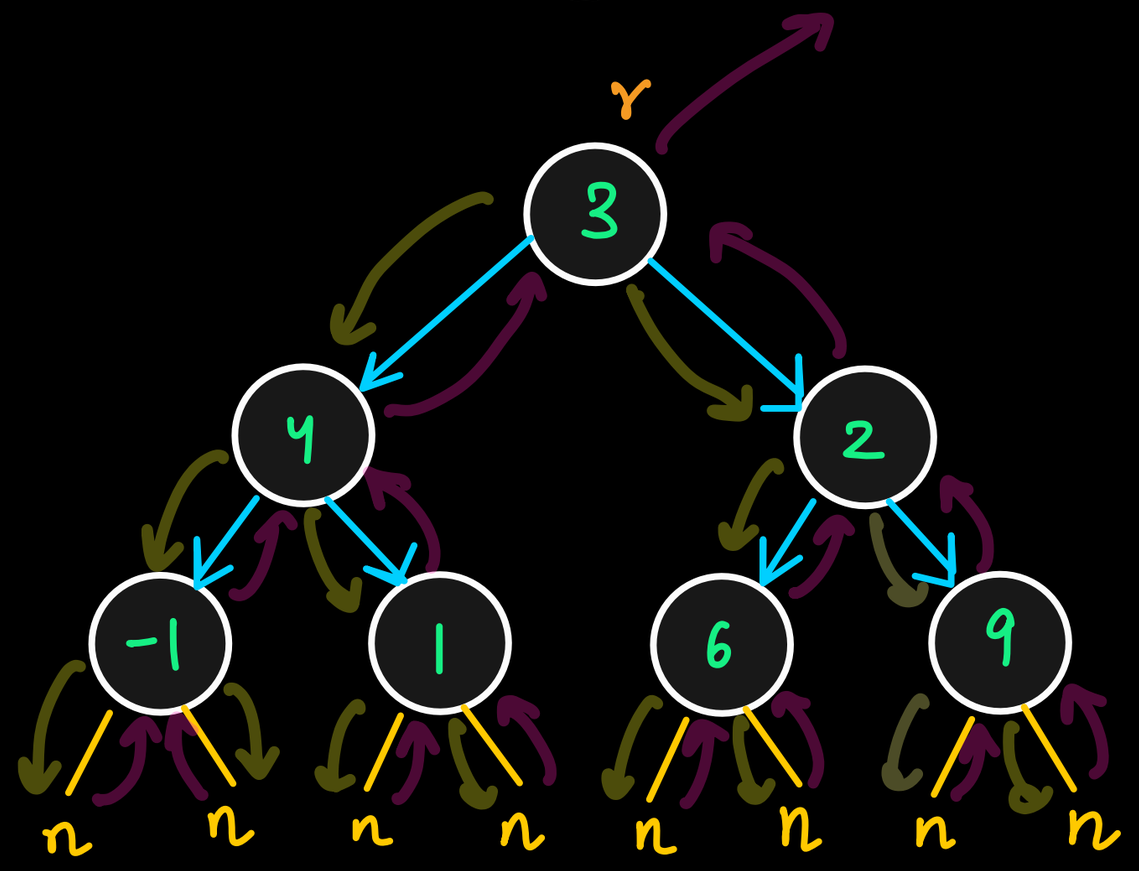
What is a Binary Tree?



Node Class

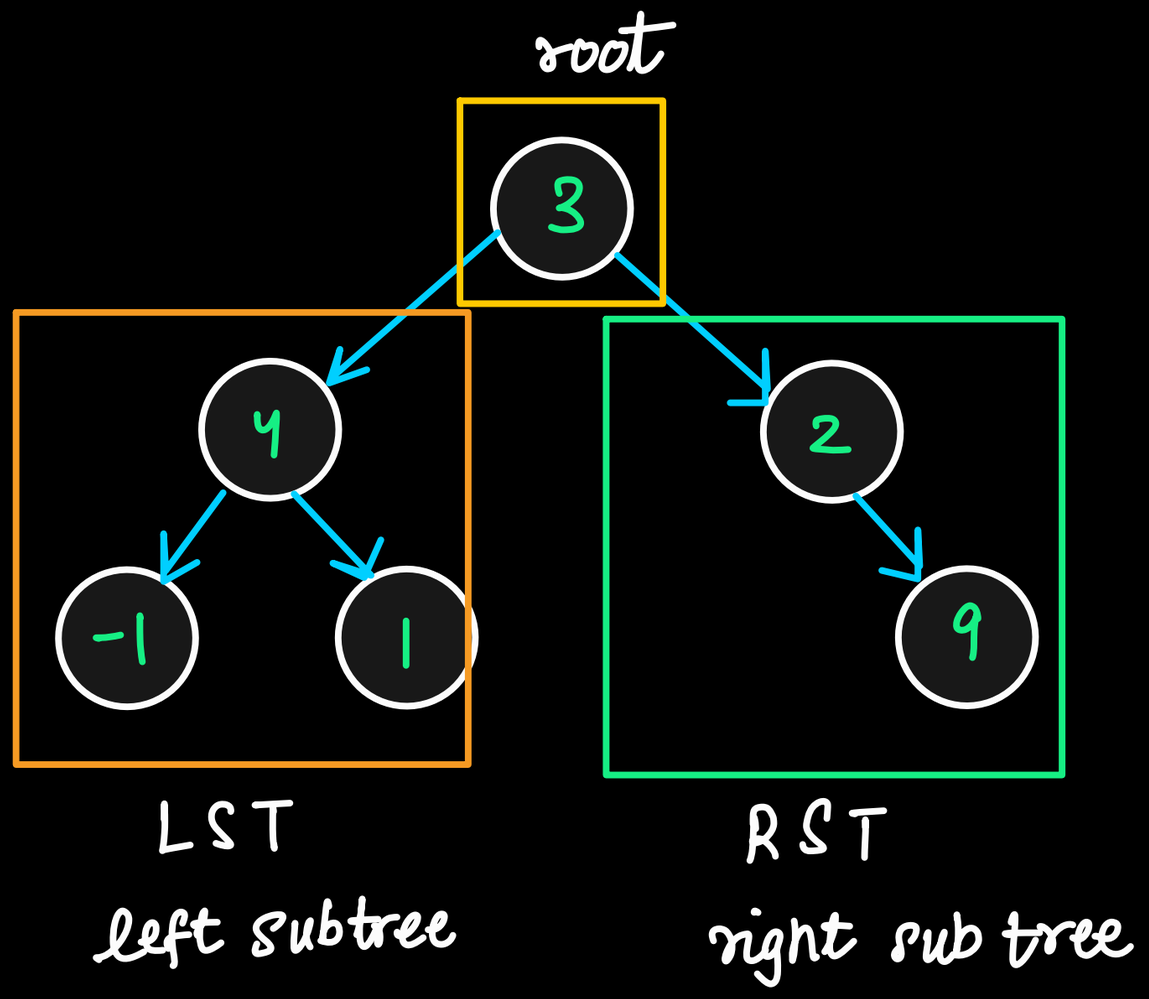
```
class Node {
    int val;
    Node left;
    Node right;
}
```

Traversing a Tree

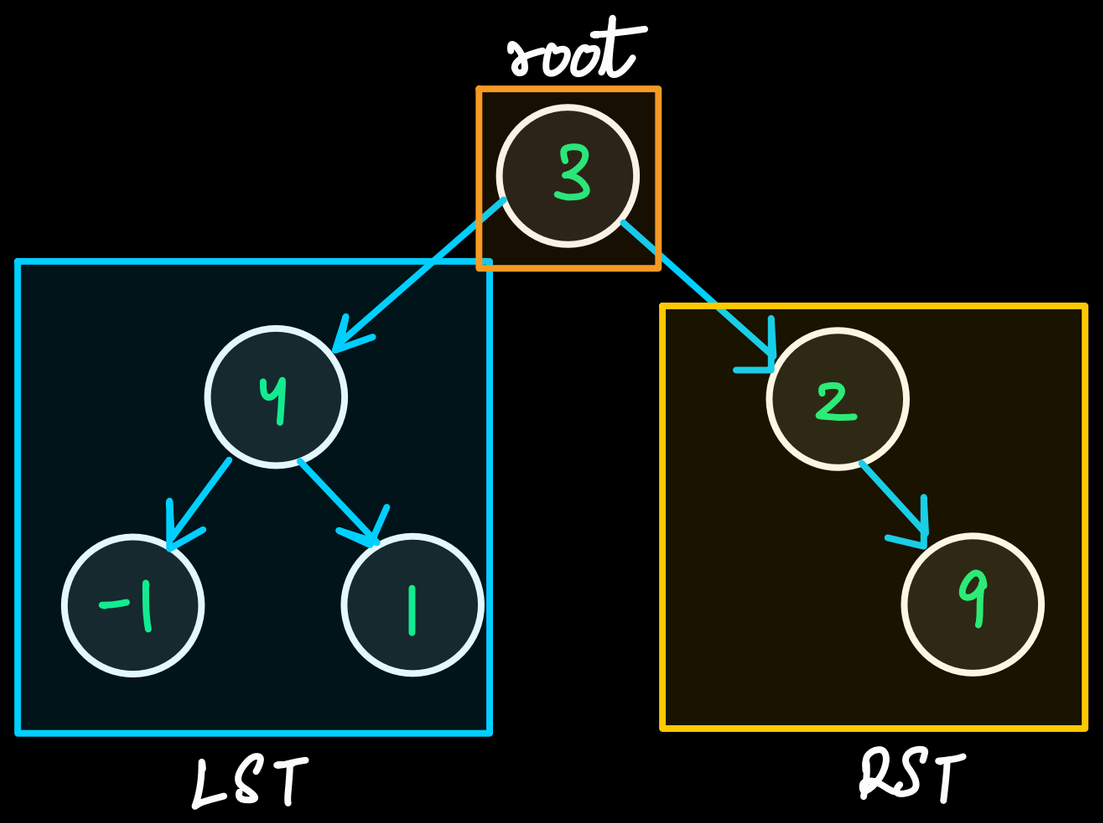



```
display(root){
    if(root==null) return;
    sout(root.val);
    display(root.left),
    display(root.right);
}
```

Output : 3 4 -1 1 2 6 9

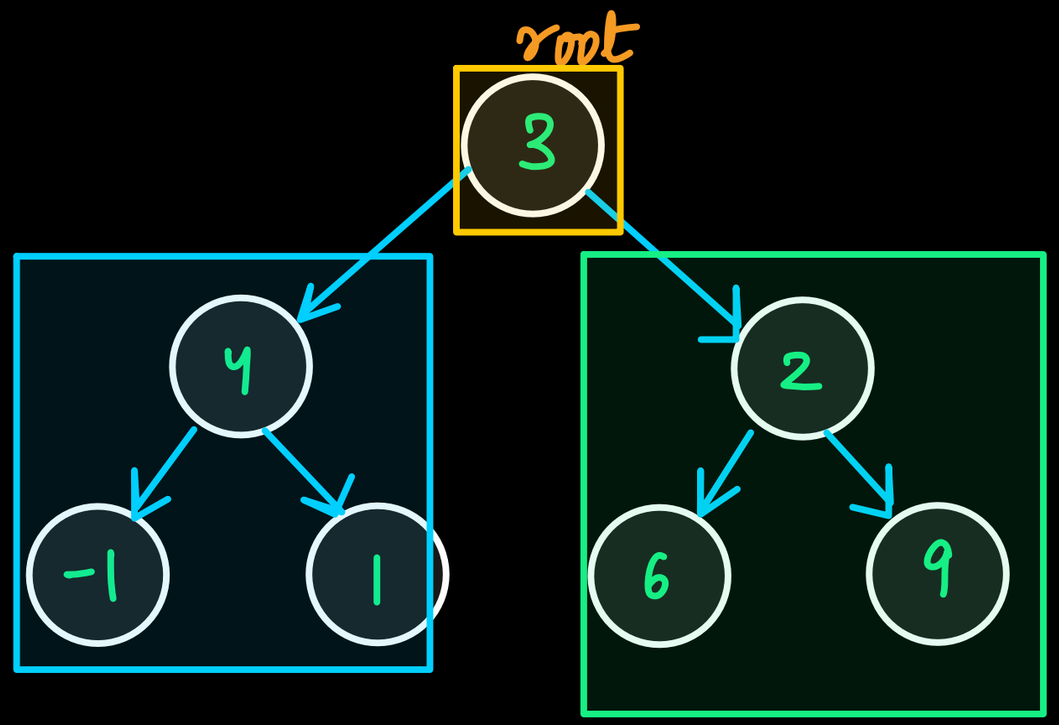

root
3
4
2
-1
1
9
LST
left subtree
RST
right sub tree

Ques: Size of a Binary Tree




```
int size(root){
    if(root==null) return 0;
    return
      1 + size(root.left) + size(root.right);
}
```

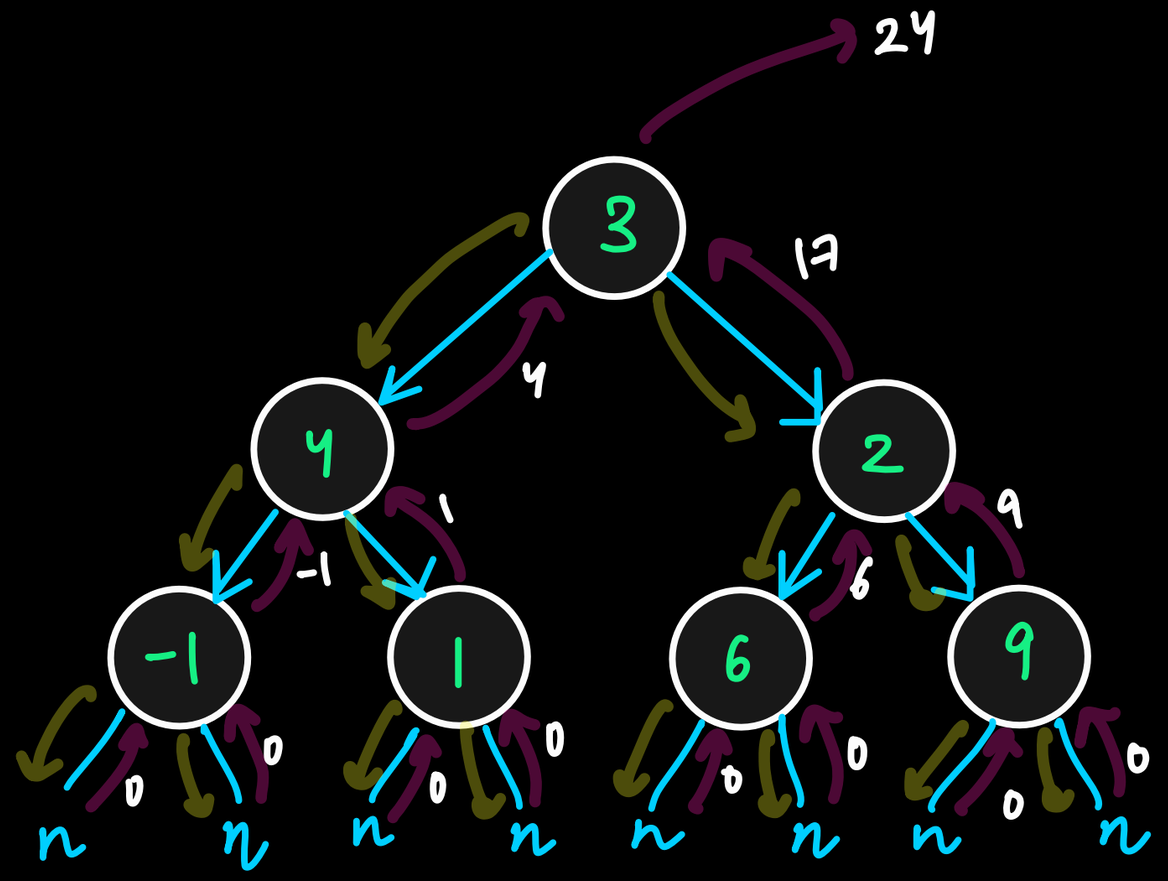
Ques: Sum of values of a Binary Tree




```
if(root == null) return 0;
```

```
sum(root) = root.val + sum(root.left) + sum(root.right);
```

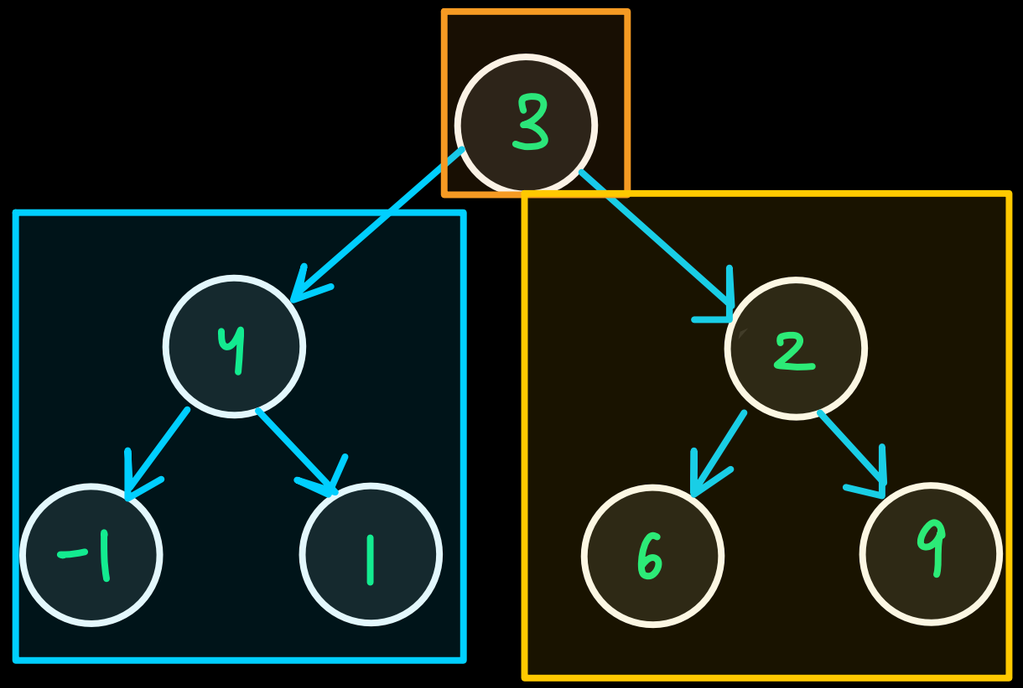
Ques: Sum of values of a Binary Tree



HW: Product of values of a Binary Tree

H.W. Product of non-zero values of a B. Tree

Ques: Maximum value in a Binary Tree




```
if(root==null)
    return Integer.MIN_VALUE
```

max = max(root.val, max(LST), max(RST))

HW: Minimum value in a Binary Tree